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Case Study - I

BrightMart is a medium sized online shopping company. Customers use its website and mobile application to purchase products.

The Company stores:

1. Customer Names
2. Email Addresses
3. Phone Numbers
4. Delivery Addresses.
5. Passwords
6. Order Information
7. Payment Related Information.

BrightMart has around 100 employees.

The company recently hired several employees. To make things easier, the IT department gave employees access to different internal systems.

However, the company has the following security weaknesses:

- Some Employees use simple passwords such as BrightMart 123.
- Many Employees use the same password for multiple systems.
- Software updates are sometimes delayed.
- The company does not regularly conduct employee cybersecurity training.
- There is no multi factor authentication for some important accounts
- Employees can access sensitive information that they do not necessarily need for their jobs.
- Backups are performed only once a month.

The Incident:
One Monday morning, an employee named Rahul receives an e-m
Subject: Urgent - Your account will be suspended.
The e-mail appears to come in the company's IT Department.
It says:
"Your company account will be suspended today. Click the link below & verify your password immediately."
Rahul clicks the link.
The page looks almost identical to the company's login page.
He enters his username & password.
Rahul does not realise that the website is controlled by an attacker.
The attacker now has Rahul's login credentials.

Stage I: Unauthorised Access.
The attacker uses Rahul's credentials to access BrightMart's internal system.
Because Rahul uses the same password for another system, the attacker is also able to access that system.
The attacker discovers that Rahul has access to customer information.
The attacker downloads a large amount of customer data.
What cybersecurity problem has occurred?

Stage II: Malware Attack.

The attacker sends another e-mail to Rahul containing a malicious attatchment
Rahul opens the attachment.
Malware is installed in his computer.
The malware begins communicating with the attacker's system.
The attacker now attempts to move from Rahul's computer to other systems within BrightMart's Network.
Which vulnerability allowed the attacker to move further into his organisation?

Stage III : Ransomware.
Two days later, employees arrive at work & discover that they cannot access their important files.
A message appears on their computers :
"Your files have been encrypted. Pay the ransom to recover them."
The company's customers database & several important business files are encrypted.
Unfortunately, the company's most recent backup is almost a month old.
BrightMart's websites stop working
Customers cannot place orders
The customer support team cannot access customer records.

Stage IV : Business Impact.
BrightMart now faces several problems.
Financial Impact -
The company loses sales because its website is unavailable.
Data Impact -
Customer information may have been stolen
Operational Impact -
Employees cannot access important systems.

Reputation Impact -
Customers lose confidence in the company.

Legal / Compliance Impact :
The company may face regulatory or legal consequences depending on what information was exposed and which laws apply.

Group I - Identify the threats
What threats are present in this case?

Group II - Identify the Vulnerabilities.
Which weakness in BrightMart's security allowed the attack to succeed?

Group III : Apply the CIA Triad
Was confidential information exposed?
Was information changed or damaged?
Were systems available to legitimate users?

Group IV : Security Solutions.
Question 1.
Who is responsible for cybersecurity?
Question 2
Was phishing the only problem?
Question 3.
What was the biggest vulnerability.

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Case Study - I - Solutions

Q. What cybersecurity problem has occured?

Ans Possible Answers:

- Phishing
- Credential theft
- Unauthorised access.
- Data Breach.

Q. Which vulnerability allowed the attacker to move further into the organisation?

Ans. Students should identify possibilities such as:

- Weak Passwords.
- Excessive user privileges
- Lack of Multi Factor Authentication
- Poor network security
- Lack of employee awareness
- Outdated software.

Q. What threats were present in this case?

Ans. They may identify:

- Phishing
- Malware
- Ransomware
- Credential Theft.
- Unauthorized Access.
- Data Theft.

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Q. What weaknesses in BrightMart's security allowed the attack to succeed?

Ans: Expected Answers:

Vulnerability	How it helped the attacker.
Weak Passwords	Easier to comprise accounts.
Password reuse	One stolen password gives access to multiple systems.
No MFA	Stolen Password may be enough to log in.
Lack of training	Employee falls for phishing.
Delayed updates	Vulnerable software may be exploited.
Excessive priviledges	Attacker can access more data.
Infrequent backups	Difficult to recover afta ransomware.

→ Confidentiality
Was confidential information exposed?
Yes.
Customer information was accessed/downloaded by the attacker.

→ Integrity
Was information changed or damaged?
Potentially Yes.
The attacker could modify or encrypt the files.

→ Availablity
Were systems available to legitimate users?
No.
The website & internal systems became unavailable because of ransomware.

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So,
Confidentiality → Affected.
Integrity → Affected
Availability → Affected.

Group 4 → Security solutions.
What should BrightMart do to prevent this from happening again?
→ Use strong passwords.
→ Implement Multi Factor Authentication (MFA)
→ Provide regular cybersecurity awareness training
→ Conduct phishing awareness exercises.
→ Keep software updated.
→ Follow the principle of least privilege.
→ Segment important networks.
→ Maintain regular backups.
→ Encrypt sensitive data.
→ Deploy antivirus & endpoint protection
→ Monitor suspicious login activity.
→ Develop an incident response plan.

Q. Who is responsible for cybersecurity?
Ans: Not only the IT department. Every employee who uses organisation's systems has a role in cybersecurity.

Q. Was phishing the only problem?
No.
Phishing was the initial attack, but weaknesses such as password reuse, lack of MFA, excessive privileges, poor backups & insufficient training allowed the attack to become much more damaging

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Q. What was the biggest vulnerability?

Ans: "Lack of employee awareness was a major vulnerability because the attack started when Rahul clicked the phishing link."

Other valid reason could be,

"Poor backup practices were the biggest problem because BrightMart could not quickly recover from ransomware".

Zuha
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